



Name: _____ Cohort/Batch: _____ Date: _____ Score: _____

WEIGHTS & BIASES PRACTICE SHEET - 10 CONCEPTUAL Q&A

10 conceptual Q&A MLOps/LLMOps

TOTAL: 10 QUESTIONS

Q1 What is Weights & Biases and what AI/ML use case is it designed for? Great Model

Q2 What are the core abstractions in Weights & Biases? Architecture

Q3 How does Weights & Biases handle training or inference? Logging

Q4 How does Weights & Biases manage data preprocessing? Preprocessing

Q5 How does Weights & Biases support GPU/TPU acceleration? IdP

Q6 How do you evaluate a model's performance in Weights & Biases? Metrics & Variability

Q7 How do you deploy a model trained with Weights & Biases? Deployment Cycle

Q8 How does Weights & Biases handle distributed or multi-GPU training? Multi-GPU Inference

Q9 How do you track experiments and versions in Weights & Biases? Testing

Q10 What are common pitfalls when using Weights & Biases in production? Monitoring



ANSWER KEY & EXPLANATORY SOLUTIONS

Comprehensive verified answers, best-practice configs, security controls & reference

VERIFIED SOLUTIONS

A1 Weights & Biases is a framework or

Mitigation

Weights & Biases is a framework or service targeting a specific AI/ML domain training neural networks, building LLM pipelines, deploying models, or orchestrating agents. Understanding its scope helps you know when to use it vs alternatives.

A6 Evaluation uses a held-out validation set and

Observability

Evaluation uses a held-out validation set and metrics (accuracy, F1, BLEU, perplexity) appropriate to the task. Monitor both training and validation metrics to detect overfitting. Use a test set only at the very end to report final results.

A2 Weights & Biases organizes work around abstractions

Primitives

Weights & Biases organizes work around abstractions like models, tensors, pipelines, agents, or embeddings. Learning these core types is the first step to using the framework effectively.

A7 Export the model to a portable format

Automation

Export the model to a portable format (ONNX, TorchScript, SavedModel). Serve it via a model server (TorchServe, TF Serving, Triton) or embed it in an API endpoint. Monitor inference latency and drift in production.

A3 For training, Weights & Biases defines a

Hardening

For training, Weights & Biases defines a model architecture, a loss function, and an optimizer. For inference, a trained model processes input data and returns predictions. Batch processing and hardware acceleration are key performance levers.

A8 Weights & Biases provides data-parallel or model-parallel

Performance

Weights & Biases provides data-parallel or model-parallel strategies to split work across GPUs. Data parallelism replicates the model on each GPU and averages gradients. Model parallelism splits layers across devices for models too large for one GPU.

A4 Weights & Biases provides data loaders, tokenizers,

Gotchas

Weights & Biases provides data loaders, tokenizers, or pipeline components that transform raw data into the tensor or embedding format the model expects. Proper preprocessing is critical for model correctness and training speed.

A9 Use an experiment tracker (MLflow, Weights &

Assessment

Use an experiment tracker (MLflow, Weights & Biases, Comet) alongside Weights & Biases to log hyperparameters, metrics, and artifacts. Track the code version and data version for full reproducibility.

A5 Weights & Biases detects available accelerators and

Federation

Weights & Biases detects available accelerators and moves computation to them. Specify the device explicitly (cuda, mps, tpu) to avoid accidentally running expensive operations on CPU. Mixed-precision (fp16/bf16) further reduces memory and increases throughput.

A10 Common issues include training-serving skew (different

Optimization

Common issues include training-serving skew (different preprocessing), missing input validation, no latency SLA enforcement, models not versioned, and no process to retrain on drifted data distributions.